

Education

Bachelor of Technology (B.Tech) in Computer Science and Engineering (CSE)

Vellore Institute of Technology, Chennai

Expected Graduation: 2026

Senior Secondary (Class XII)

Vivekanand Public School

July 2020 – 2022

Experience

Apoliums Infotech India Pvt. Ltd.

Dec 2025 – Jan 2026

Backend Engineering Intern

- Migrated legacy Node.js services to **Golang (Gin)** and optimized MySQL schemas, reducing API latency and enhancing concurrency via clean architecture.
- Engineered high-performance **RESTful APIs** deployed on **GCP (Compute Engine, Cloud SQL)**, ensuring robust system reliability and seamless production operations.

Research Intern

May 2025 – July 2025

Centre for e-Automation Technologies (CeAT), VIT Chennai

- Developed a **CloudSim Plus** framework to evaluate the **Hippopotamus Optimization (HO)** algorithm for VM placement in cloud data centers.
- Enhanced simulation scalability by modularizing components, enabling reproducible testing across multiple datacenter configurations —GitHub: Repo

Full Stack Developer Intern

Dec 2024 – Feb 2025

Daira Edtech Pvt Limited

- **Architected the Activity Management System** using Node.js and MongoDB, engineering RBAC with secure **JWT** flows to facilitate project lifecycles for faculty and external reviewers.
- **Enhanced system reliability** and Student Progress tracking by refactoring legacy logic and implementing rigorous server-side testing, significantly reducing API errors and downtime.

Web Development Intern

Sept 2024 - Oct 2024

Indian Institute of Technology Bombay (IIT Bombay)

- Reduced data payload size by 90% for low-bandwidth environments by replacing text messages with numerical status codes in JSON responses.
- Enhanced backend robustness and API reliability by implementing Joi schema validation for the SaaS platform.

Projects

Hybrid Ensemble Learning for Early Abiotic Stress Detection in Crops

Aug 2025 – Nov 2025

- Developed a hybrid ensemble pipeline (LSTM w/ Attention + XGBoost) on 30-day sensor data, achieving 99.78% accuracy and 0.954 Macro F1 by learning robust temporal representations.
- Optimized performance on limited datasets using an 8-timestep sliding window to generate 64-dimensional temporal embeddings, ensuring reliable early-stage abiotic stress detection.

Technologies: Python, LSTM, Attention Mechanisms, XGBoost, Scikit-learn, Pandas, Time-Series Analysis

High-Performance Parallel AES Encryption Framework

Jan 2025 - Apr 2025

- Reduced encryption time by 51% on large payloads by engineering parallelized AES via Python’s ProcessPoolExecutor.
- Increased throughput by 21.9% (to 362 KB/sec) vs. standard AES by designing a multi-core CTR-mode pipeline.
- Achieved a 49.95% avalanche effect, confirming cryptographic robustness of the dynamic S-box design.

Technologies Used: Python, Multiprocessing, SHA-512, Matplotlib, NumPy

Water Brakes

Aug 2024 - Nov 2024

- Developed a Streamlit app analyzing contour maps to recommend precise swale and trench locations for water management.
- Enabled data-driven groundwater recharge decisions by processing UTM data to generate interactive Plotly visualizations.

Technologies Used: Streamlit, Python, Plotly, Matplotlib | Live: water-brakes.streamlit.app

Skills

- **Programming Languages:** Python, C++, C, Java, HTML, CSS, SQL, JavaScript, Bash.
- **Databases & Cloud:** MongoDB, PostgreSQL, Amazon Web Services (AWS), Google Cloud (GCP), Docker.
- **Tools & Frameworks:** Git, React, Next.js, Node.js, Express.js, RESTful APIs, Linux.
- **Core Concepts:** Data Structures & Algorithms, DBMS, Operating Systems, Continuous Integration/Continuous Development.

Positions of Responsibility

Outreach Lead, DAO Community:

- Grew community membership by 50% in 6 months through structured outreach and on-campus workshops in collaboration with Web3 companies including Polygon, Spheron, Huddle01, blockOn, and wazirX.
- Secured partnerships for 15+ national hackathons and TOKEN2049 Singapore; led organization of 5 large-scale hackathons and managed a 35-member team for Defy24, achieving record sponsorship and profit.

Blockchain Lead, Hackclub:

- Scaled the blockchain department by 300% in 6 months, recruiting and mentoring 25+ junior developers.
- Designed and delivered blockchain workshops, learning roadmaps, and personalized mentorship to upskill members.